



Airbus SPOT: Quality Processes Review

Author(s): ***Enis Gerxhalija***
National Physical Laboratory

Reviewer(s): ***Sam Hunt, Samantha Malone***
National Physical Laboratory

Contents

1	Introduction	2
1.1	Scope	2
1.2	Changes Log.....	2
1.3	Applicable Documents.....	2
1.4	Reference Documents	2
1.5	Acronyms & Abbreviations.....	3
2	Review Summary – Product Maturity Matrix	4
3	Product Information	5
4	Metrology.....	9
5	Product Generation	12

1 Introduction

1.1 Scope

This document summarises a Data Product Quality Processes Review for Airbus SPOT as part of the EODH Data Product Quality Process [AD-1]. The purpose is to demonstrate that the product has been developed and generated following processes that adhere to best practices.

To achieve this, the EODH Data Product QA Process adheres to the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects of Data Product quality – awarding grades of *Basic*, *Good*, *Excellent*, or *Ideal*. Such assessments are reported with a maturity matrix – this is presented in Section 2. The rest of the document provides the underpinning evidence justifying the awarded grades. This assessment has been completed by the EODH QA Service on behalf of the Data Product provider.

The result of this evaluation, referred to as a *QA Check Result*, will be added as an annotation to the EODH Catalogue for user interaction.

1.2 Changes Log

Issue	Date	Description
0.1	03/11/2025	Version for internal review
1.0	20/02/2026	Draft for pre-publication review.

1.3 Applicable Documents

[AD-1] EODH QA Service, 'Data Product Quality Assurance Process', 2025.

1.4 Reference Documents

[RD-1] Hunt et al., 'Earth Observation Mission Quality Assessment Framework' v2.2, EDAP Project Document, 2021. [Available Online](#).

[RD-2] Airbus SPOT Imagery User Guide. [Available Online](#)

[RD-3] Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci Data 3, 160018 (2016); <https://doi.org/10.1038/sdata.2016.18>

[RD-4] JCGM, "Evaluation of measurement data - Guide to the expression of uncertainty in measurement," *JCGM*, vol. 100, 2008, Accessed: Feb. 06, 2018. [\[Online\]](#)

1.5 Acronyms & Abbreviations

EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration
NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory
QA	Quality Assurance

2 Review Summary – Product Maturity Matrix

The *Data Product Quality Processes Review* is aimed at demonstrating that a given Data Product is developed and generated following processes that adhere to best practices.

To achieve this, the EODH *Data Product QA Process* adheres to the ESA/NASA EO Product QA Framework [RD-1]. Such assessments are reported in a maturity matrix style, where each box represents an area of assessment for which underpinning requirements are defined – the box colour represents the achieved grade, *Basic*, *Good*, *Excellent*, or *Ideal*. Additionally:

- *Not Assessed* means this aspect of data quality was not assessed at this time.
- Not Assessable means the assessment of this aspect of data quality was not possible due to a lack of available evidence.
- The padlock symbol means that the supporting evidence reviewed is not publicly available.

The *Data Product Quality Processes Review* for Airbus SPOT is presented in Figure 1. The rest of the document provides the underpinning evidence justifying the awarded grades, as follows:

- Section 3 provides the evidence for the Product Information column.
- Section 4 provides the evidence for the Metrology column.
- Section 5 provides the evidence for the Product Generation column.

Data Product Quality Processes Review			Key
Product Information	Metrology	Product Generation	
Product Details	Sensor Calibration & Characterisation	Calibration Algorithm	Not Assessed
Availability & Accessibility	Geometric Calibration & Characterisation	Geometric Processing	Not Assessable
Product Format, Flags & Metadata	Metrological Traceability Documentation		Basic
User Documentation	Uncertainty Characterisation		Good
	Ancillary Data		Excellent
			Ideal
			Not Public

Figure 1 - Data Product Quality Processes Review Maturity Matrix for Airbus SPOT

3 Product Information

Product Details	
Grade: Excellent	
Justification	<i>Almost all required information available publicly, located in the user guide [RD-2]</i>
Product Name	<i>Airbus SPOT MS ORT</i>
Sensor Name	<i>New AstroSat Optical Modular Instrument (NAOMI)</i>
Sensor Type	<i>Panchromatic & Multispectral (Blue, Green, Red, NIR) Array Assembly</i>
Mission Type	<i>Constellation – 2 twin satellites (SPOT 6 & 7).</i>
Mission Orbit	<i>Sun-synchronous; 10:00 AM local time at descending node, 26-day cycle, 694-km altitude</i>
Product Version Number	<i>Included in product metadata</i>
Product ID	<i>Product name for Primary and Ortho files:</i> <i>SPOT<SAT_NUMBER>_<SPECTRAL_PROCESSING>_<IMAGING_TIME>_<PROCESS_LEVEL>_<JOB_ID></i> Example: <i>SPOT6_PMS_201006181052299_ORT_123456789</i> <i>Name(s) of SPOT 6/7 Data Strip Source metadata file(s) is comprised of:</i> <i>DS_POT<n>_<YYYYMMDDHHMMSS>_<AAA>_<aaa>_<RRR>_<rrr>_<XxxxYyy>_<TTTT></i> <i>Example:</i> <i>STRIP_DS_SPOT6_201212051035424_FR1_FR1_FR1_FR1_E002N41_01174</i> - <i><YYYYMMDDHHMMSS> UTC strip source acquisition start time</i> - <i><AAA> = effective archiving center acronym</i> - <i><aaa> = planned archiving center acronym</i> - <i><RRR> = effective receiving center acronym</i> - <i><rrr> = planned receiving center acronym</i> - <i><XxxxYyy> = closest square degree of the strip center with:</i> - <i>X = {W,E}, West or East, xxx longitude degree (000 to 180)</i> - <i>Y = {N,S}, North or South, yy latitude degree (00 to 90)</i> - <i><TTTT> = Number of PAN source frames</i>
Processing level of product	<i>Primary Product</i> <i>Standard Ortho Product</i> <i>Tailored Ortho Product</i>
Measured Quantity Name	<i>TOA Radiance</i> <i>TOA Reflectance</i>

Measured Quantity Units	<i>TOA Radiance – $W\ sr^{-1}\ m^{-2}\ \mu m^{-1}$</i>
Stated Measurement Quality	<i>Absolute calibration- <5% except NIR (<10%) Relative cross-band calibration- 3% Relative multi-temporal calibration- 5%</i>
Spatial Resolution	<i>Panchromatic: 1.5m Multispectral: 6m</i>
Spatial Coverage	<i>6 million km² per day, 60 km image swath. Up to 45° viewing angle.</i>
Temporal Resolution	<i>Dependent on viewing angle and simultaneous operation</i> - 1 day with SPOT 6/7 operating simultaneously. - Up to 3 days with only one in operation (dependent on latitude).
Temporal Coverage	<i>SPOT 6- September 2012 – still operational at time of report SPOT 7- June 2014 – March 2023</i>
Point of Contact	<i>Airbus DS Intelligence Customer Care</i>
Product locator (DOI/URL)	<i>Available for purchase on the Airbus SPOT Website</i>
Conditions for access and use	<i>Requires commercial licensing</i>
Limitations on public access	<i>Yes (see above).</i>
Product Abstract	<i>With SPOT 6 and SPOT 7, Airbus Defence and Space (DS) not only secure mission continuity of its SPOT series, which has collected an archive of more than 30 million scenes since 1986.</i> <i>This new generation of optical satellites features technological improvements and advanced system performance that increase reactivity and acquisition capacity, as well as simplifying data access. The first ‘twins’ in the SPOT family, SPOT 6 and SPOT 7, are 180-degree phased, enabling a revisit at least once a day for any point on Earth.</i>

Availability & Accessibility	
Grade: Good	
Justification	<i>Good availability & accessibility with dataset meeting many FAIR principles [RD-3]. Detailed overview of product content structure, metadata and flags provided.</i> <i>Unable to find data management plan.</i>
Compliant with FAIR principles	No
Data Management Plan	Not known
Availability Status	Commercially available via website , phone, email or fax.

Product Format, Flags & Metadata	
Grade: Excellent	
Justification	<i>DIMAP V2, a documented standard file format, is used. The product includes a well-detailed set of metadata & flags, but not CEOS-ARD compliant.</i>
Product File Format	<i>Products are available in a well-defined file format structure defined in the Appendix of [RD-2]:</i> <i>Users can specify the product and image format, from:</i> • DIMAP – JPEG 2000 Optimised • DIMAP – JPEG 2000 Regular • DIMAP – GeoTIFF • NITF – JPEG 2000 Regular • NITF – GeoTIFF <i>DIMAP V2 is the default Pléiades product format. Inside that product, you can select an image format:</i> • JPEG 2000: regular compression (8 bits/pixels) • JPEG 2000: optimised compression (3.5 bits/pixels), • GeoTIFF: uncompressed.
Metadata Conventions	Default DIMAP V2 format
Analysis Ready Data?	No

User Documentation		
Grade: Good		
Justification	<i>There is currently a user guide (see [RD-2]), but there is no ATBD made publicly available. However, there is ATBD-like information in the user guide for geometric modelling, and spectral modelling.</i>	
<i>Document</i>	<i>Reference</i>	<i>QA4ECV Compliant</i>
Product User Guide	<i>Airbus SPOT Imagery User Guide [RD-2]</i>	<i>No</i>
ATBD	<i>Documentation not made available to users.</i>	<i>N/A</i>

4 Metrology

Sensor Calibration & Characterisation	
Grade: Basic	
Justification	<i>Any formal sensor calibration and characterisation documentation (for either pre-flight or post-launch) is not available publicly.</i> <i>There is some very high-level information in [RD-2] which outlines the main radiometric corrections/enhancements that have been carried out:</i> <i>• Inter-detector equalisation: correction of differences in</i> <i>• sensitivity between the detectors (on-board correction).</i> <i>• Aberrant detectors correction (none at that time).</i> <i>• Panchromatic band restored and de-noised.</i> <i>• Pixel sampling at Shannon optimizing image quality for downstream value-added processing: optimized bicubic kernel resampling into the Primary geometry, zoomed to the factor 7 / 5 (equivalent resolution of 1.5/6m in nadir condition)</i> <i>Further details on the methods used to evaluate these corrections is not available publicly.</i>
References	[RD-2]

Geometric Calibration & Characterisation	
Grade: Basic	
Justification	<i>Any formal documentation describing geometric calibration is not available publicly.</i> <i>ATBD-type information on the geometric modelling is provided in the Appendix [RD-2].</i> <i>There is also high-level information that outlines the main geometric processing:</i> <i>• The combination of all sub-swaths across in the field of view (60 km nadir condition): synthesis in a virtual focal plane represented by a single linear array for all spectral bands.</i> <i>• Correction of instrumental and optical distortions: viewing angles adjusted to the single linear array model</i> <i>• Co-registration of all spectral bands: Multispectral and Panchromatic</i> <i>• Attitudes and ephemeris data are refined at ground on the mean estimation:</i>

	○ Adjustment on ground reference data if available ○ Attitudes filtering over time of acquisition • Consistent alignment of the physical model ancillary data and RPC analytic model data
References	[RD-2]

Metrological Traceability Documentation	
Grade: Not Assessable	
Justification	<i>There are no metrology documentation or traceability chain / uncertainty tree diagrams publicly available.</i>
References	[RD-2]

Uncertainty Characterisation	
Grade: Not Assessable	
Justification	<i>Information is given in Appendix B of the user guide [RD-2] on the performance regarding Geometric, Radiometric, and Resolution performance.</i> <i>Reference to some pre-flight calibration in the Radiometric performance characterisation, but no comparison to other sensors made, and doesn't follow GUM approach [RD-4].</i>
References	[RD-2], [RD-4]

Ancillary Data	
Grade: Good	
Justification	<i>Overview of the ancillary data for the geometric details of the orthorectified product is available in the appendix of the user guide [RD-2]:</i> • Geographic projections: WGS84 – latitude/longitude sampling according to ARC or DTED2 zoning standards (please refer to A.5.1 for more details). • Mapping projection: Most of the projections registered by EPSG, in metres (please refer to A.5.2 for more details). • GCP: Reference3D Ortho layer. • DEM: Reference3D DEM layer (DTED2), SRTM (DTED1), GLOBE (DTED0) <i>No mention of ancillary data uncertainty.</i> <i>An RPC XML file (containing the coefficients and Normalisation parameters for the Rational Polynomial Coefficients (also called Rapid Positional Capability, Rational Function Model) geometric</i>

	<i>analytic model) is also provided which allows users to carry out geometric processing with software that supports RPC models.</i>
References	<i>[RD-2]</i>

5 Product Generation

Calibration Algorithm	
Grade: Not Assessable	
Justification	<i>No information made available publicly.</i>
References	

Geometric Processing	
Grade: Good	
Justification	<i>ATBD-like information on the geometric processing given in the appendix of the user guide with description of input parameters and their tags in the XML metadata file.</i>
References	<i>[RD-2]</i>